```
sit@sit-OptiPlex-3280-AIO: ~/Music/1086_cc/NineExp — ./a.out
~/Music/1086_cc/NineExp

sit@sit-OptiPlex-3280-AIO:~/Music/1086_cc/NineExp$ yacc -d MyNineth.y
sit@sit-OptiPlex-3280-AIO:~/Music/1086_cc/NineExp$ flex MyNineth.l
sit@sit-OptiPlex-3280-AIO:~/Music/1086_cc/NineExp$ gcc y.tab.c lex.yy.c -lm
^[[Asit@sit-OptiPlex-3280-AIO:~/Music/1086_cc/NineExp$ ./a.out
 SDT DESK CALCULATOR
Operators: + - * / % ^ ++ --
Press Ctrl+D to exit.

4 % 4 / 0
Error: Division by zero
Result = 0
6++ - 10 / 2
Error: syntax error
Error: Invalid expression. Please try again.
++6 - 10 / 5
Result = 5

8-2 ^ 4 + 6++
Error: syntax error
Error: Invalid expression. Please try again.
6++
Error: syntax error
Error: Invalid expression. Please try again.
25/5 - 3^4 + 5
Result = -71
```

Name: Parinita Choudhari

PRN: 24070521086

Batch: B1

Date of Execution: 22/09/26

Sep 22 12:20

Open

MyNineth.y
~/Music/1086_cc/NineExp

MyNineth.y

```
%{
#include <stdio.h>
#include <math.h>

#define YYSTYPE double

int yylex(void);
void yyerror(const char *s);
%}

%token NUMBER
%token INC DEC
%token INVALID

%left '+' '-'
%left '*' '/' '%'
%right '^'

%%

input:
      /* empty */
    | input line
    ;

line:
      expr '\n'
        {
            printf("Result = %g\n", $1);
        }
    | '\n'
        {
        }
    | error '\n'
        {
            printf("Error: Invalid expression. Please try again.\n");
            yyerrok;
        }
    ;
```

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Open

MyNineth.y
~/Music/1086_cc/NineExp

MyNineth.y

```
48        {
49            $$ = $1 + $3;
50        }
51
52    | expr '-' expr
53        {
54            $$ = $1 - $3;
55        }
56
57    | expr '*' expr
58        {
59            $$ = $1 * $3;
60        }
61
62    | expr '/' expr
63        {
64            if ($3 == 0)
65            {
66                yyerror("Division by zero");
67                $$ = 0;
68            }
69            else
70            {
71                $$ = $1 / $3;
72            }
73        }
74
75    | expr '%' expr
76        {
77            if ($3 == 0)
78            {
79                yyerror("Modulus by zero");
80                $$ = 0;
81            }
82            else
83            {
84                $$ = fmod($1, $3);
85            }
86        }
87
88    | expr '^' expr
89        {
90            $$ = pow($1, $3);
91        }
92
93    | INC NUMBER
94        {
95            $$ = $2 + 1;
96        }
97
98    | DEC NUMBER
99        {
```

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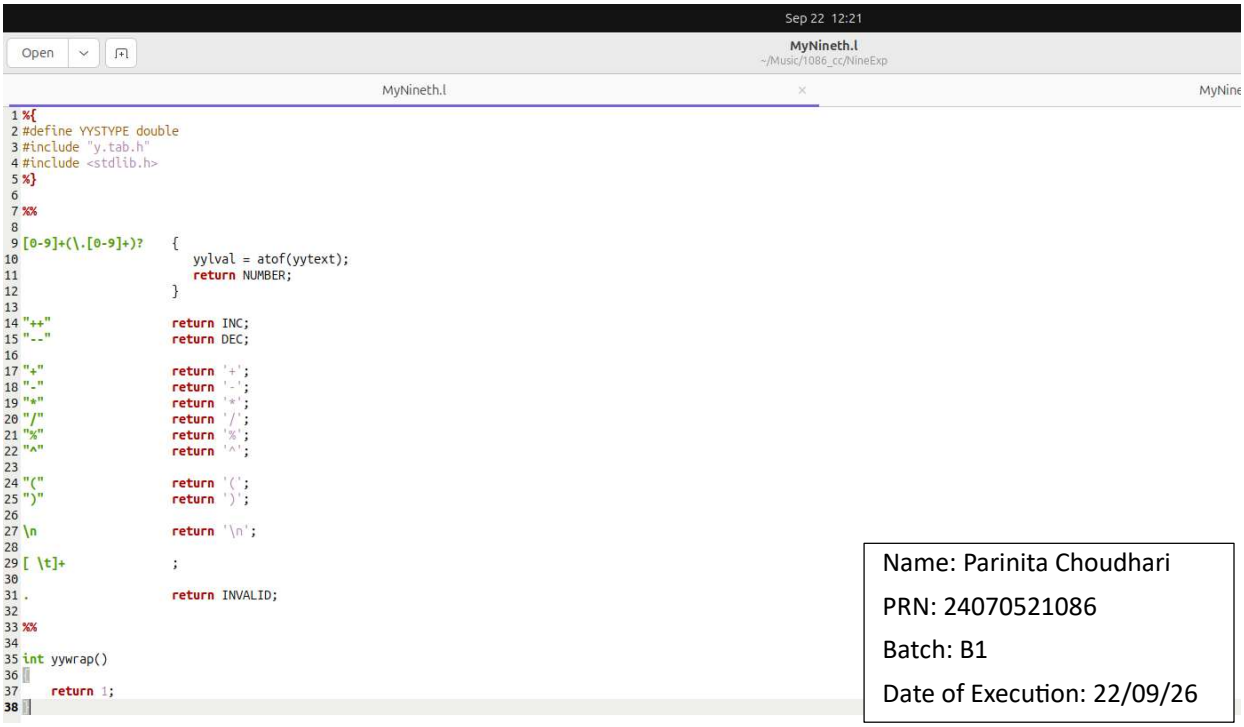
```
92
93    | INC NUMBER
94        {
95            $$ = $2 + 1;
96        }
97
98    | DEC NUMBER
99        {
100           $$ = $2 - 1;
101       }
102
103   | '(' expr ')'
104       {
105           $$ = $2;
106       }
107   ;
108
109 %%
110
111 void yyerror(const char *s)
112 {
113     printf("Error: %s\n", s);
114 }
115
116 int main()
117 {
118     printf(" SDT DESK CALCULATOR \n");
119     printf("Operators: + - * / %% ^ ++ --\n");
120     printf("Press Ctrl+D to exit.\n\n");
121
122     yyparse();
123
124     return 0;
125 }
```

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